

CHE 205, Fall 2019, Problem Set 7

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Due date: 11:59pm, Sun, 11/03/2019

All problems are from the textbook by V.L. Law.
Submit two files, one for each problem, on blackboard.

Problem 1 (6 pts). Exercise 6.2. Use forward Euler method. Compare $C_A(x)$ vs x for three different values of $\Delta x=0.1$ (black), 0.01 (red), and 0.001 (blue).

Problem 2 (6 pts). Exercise 6.3. Set $\Delta y=0.1$.

Exercise 6.2: Consider the problem of diffusion and reaction in a cylindrical pore (e.g., in a solid catalyst) where component A reacts at the walls of the cylinder according to



where

$$r = kC_A^2 \text{ (second-order reaction)} \quad (6.23)$$

In this system, component A diffuses into the pore due to lower concentration of A inside the pore than at the pore mouth. Since B is produced by the reaction, the concentration of B inside the pore is larger than at the inlet, causing diffusion of B out of the pore. At the inlet of the pore ($x = 0$), the concentration is C_{A0} . The end of the pore ($x = L$) is assumed to be sealed, so there is no flux of A at $x = L$. The mathematical model for this system can be expressed as follows:

$$\begin{aligned} D_A \frac{d^2 C_A}{dx^2} &= kC_A^2 \\ C_A(0) &= C_{A0} \\ \frac{dC_A(L)}{dx} &= 0 \end{aligned} \quad (6.24)$$

The second boundary condition is the “no flux at $x = L$ ” condition. This is a split BVP that can be solved using the shooting method. Here are some data for the problem:

$$\begin{aligned} k &= 0.01 \text{ L/(gmol s) (rate constant)} \\ C_{A0} &= 1.0 \text{ gmol/L (inlet concentration)} \\ D_A &= 1 \times 10^{-3} \text{ cm}^2/\text{s (diffusivity)} \\ L &= 1 \text{ cm (length of pore)} \end{aligned}$$

Prepare an Excel spreadsheet to solve this problem using the Euler method. Use Goal Seek to implement the shooting method. Be sure to get a reasonable value of the unknown boundary condition (by experimentation) before invoking Goal Seek.

Exercise 6.3: Consider the same problem as in Exercise 6.2, but with a first-order reaction:



where

$$r = kC_A \text{ (first-order reaction)} \quad (6.26)$$

The model equations then become

$$\begin{aligned} D_A \frac{d^2 C_A}{dx^2} &= kC_A \\ C_A(0) &= C_{A0} \\ \frac{dC_A(L)}{dx} &= 0 \end{aligned} \quad (6.27)$$

Apply the finite difference method to solve this problem (first introduce a dimensionless distance $y = x/L$). Use the same data as for the second-order reaction case. For the right-hand boundary condition, do not forget to use a second-order correct finite difference form.